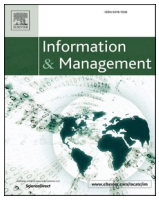
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Better job application systems: Objectively assessing measures of job performance from asynchronous video interviews

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ABSTRACT

When selecting top candidates for a job, organizations would prefer to not accidentally filter out the highest quality candidates. But an unbiased, detailed assessment of every applicant in a large candidate pool has been prohibitively costly. Asynchronous video interviews (AVIs) are inspiring new ideas for predicting job performance early in the hiring process by providing a rich source of signals. We propose that automatic analysis of interview data can improve candidate filtering at the application stage. The potential of this approach is clear, but there is a need for a structured framework and benchmarks to develop effective and valid application systems. We therefore propose a design framework that enhances the objectiveness of automated candidate assessment using AVIs through principles such as using behavioral cues that are hard to fake and using unbiased, validated labels in training sets. We demonstrate the implementation of this framework and evaluate its potential by building a prototype for automatically assessing general mental ability, an important and generalizable indicator of job performance. Results show that if new application systems adhere to this framework, more objective measures of job performance can be assessed automatically from AVI recordings. More generally, the study guides advancement of automated AVI platforms with a focus on efficacy and fairness.

1. Introduction

In 2009, Brian Acton was passed over for a job at Facebook. He went on to create WhatsApp, which he sold to Facebook in 2014 for $19 billion. In the late 1980s, Jack Ma was rejected by every company he applied to out of college, including one company that hired the other 23 candidates who applied. Ma's private equity firm now owns some of that same company. While there is no public data tracking the frequency of overlooked top candidates, anecdotal stories like Acton's and Ma's are not rare. Companies fear letting "another 'Brian Acton' slip away" [1].

However, the risk of overlooking a "Brian Acton" is growing. Technological advances have made applying for jobs easier but have also made finding the best candidate harder. Most processes still rely on a manual review of job applications to winnow down the pool to the best few. Automated solutions have been and will continue to be proposed, but these solutions can suffer from or even amplify the same bias as manual processes [2]. Further, the applications often miss or poorly represent important candidate qualifications, and this poor signal quality leaves potential to exclude the best candidates. This problem is compounded as the volume of applications increase, which increases the risk of excluding the best candidates. More accurate and unbiased automated screening processes are needed for fair and effective evaluations. These considerations inspire the research question: *How should application systems be specified to provide better insight into candidates?*

There are compelling reasons for information systems (IS) research to identify a framework for high-volume job application systems. The first is the increasing adoption of asynchronous video interviews (AVIs) as part of the application process, which provide new signals from applicants. Past IS and related research has extracted behavioral "cues" from AVIs and similar interviewing systems to predict deception [3–5]. This suggests there may also be potential for automatically estimating signals indicative of job fitness, such as intelligence and social skills. Related research [6] has automatically extracted some verbal and lexical cues from recorded face-to-face interviews and correlated them with expert assessments of job fitness factors. However, AVIs are fundamentally different from face-to-face interviews: AVI interactions are typically short, one-way responses to a camera with no follow-up questions, a difference that significantly affects both candidate

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perceptions [7] and human evaluations of candidates [8]. It is essential to thoroughly review the entire AVI process—from question formulation and behavioral cue extraction to the evaluation of interviewee characteristics and model training methods—to ensure maximum validity and fairness.

Both industry and academia have made significant strides in automating AVIs. With the rise of artificial intelligence (AI) in handling large-scale data analysis, the trend toward using AI-driven systems for screening interview responses is set to continue. However, the rapid pace of developing AI-driven systems is outstripping the evidence-based practices associated with traditional assessment tools [9]. To ensure reliability and validity, it is imperative that we establish a framework for robust application systems in the nascent phase of AVI-based systems. Before AI enters this domain in earnest, now is the ideal time to establish a robust pattern or framework to which future designs can be compared. We thus determined to outline and test a set of fundamental design principles that specify how AVI-based job-application systems can be designed to objectively measure job fitness.

1.1. Capturing signals from AVIs

This study follows a pattern of prior IS research extending *signalling theory* to socio-technical challenges, which has been central to most personnel recruitment and selection research since the 1980s. This theory describes recruitment as a signal exchange between applicants and firms, wherein job-fit signals are gleaned from applications and subsequent evaluations [10].

However, *costs* limit signalling. A thorough evaluation of hundreds of job applicants would incur substantial cost and would also deter qualified candidates who view the process as excessively time-consuming [11]. Instead, high-volume selection processes typically perform an initial winnowing through inexpensively obtained résumés and applications, which have been shown to provide somewhat unreliable signals [12,13]. And even when accurate, applications are not designed to effectively signal many important characteristics which can be obtained from extended interviews, skills exams, and other helpful but costly assessments [14,15].

We considered that if more reliable signals of candidate fit can be assessed at the application phase, using AVIs to minimize cost, then the most qualified candidates could be correctly identified more often in the earliest stages of the hiring process. The stage is set: The use of AVIs for interviews is increasing [2], and researchers are actively exploring how to automatically extract signals from video responses [see Table 1]. However, there are three challenges we identified in this nascent research domain that are addressed by the framework we will propose. First, research in this domain has predominantly used human perceptions of interviewees as the outcome variable of interest. While there is value in predicting subjective human perceptions [16], such perceptions have potential to introduce bias and even legal liability [17].

This is particularly troubling when automated tools are used in a manner that amplifies and perpetuates biases. Automated keyword matching [e.g., 18] and large-language models standardize any biases included in their training data [19]. Using these approaches as filtering tools for job applications may cut costs but increase the risk of rejecting the most qualified applicants due to systematized bias.

Second, existing research in automated signal analysis has largely overlooked signals most strongly correlated with job performance, instead focusing on aspects such as personality or communication skills. While these attributes are important for successful employees, other traits like intelligence and job knowledge may be better predictors of performance for many job types. Thus, it remains unclear whether or to what extent non-interactive AVI responses effectively signal strong job-performance indicators as measured more objectively (e.g., by an intelligence test).

Third, many AVI studies tend to neglect elements from established assessment methodologies, like the significance of question types and specific behavioral markers that are shown to enhance validity and diminish bias.

Table 1
Automated AVI Assessment Literature.

Reference	Construct(s)	Ground Truth	Modality
[54]	Soft skills, including conscientiousness, gregariousness, imaginativeness, independence, interpersonal sensitivity, perseverance, persuasiveness, social skills	Human perception	Linguistics Voice Face
[55]	Competence	Human perception	Linguistics
[56]	Personality	Self-report & human perception	Linguistics Voice Face
[57]	Competence	Human perception (supervisor)	Face
[53]	Communication skills, personality	Human perception	Face
[3]	Deception	Experimental manipulation	Linguistics Voice Face
[58]	Hireability	Human perception	Linguistics Voice Face
[59]	Hireability	Human perception	Face
[52]	Hireability, personality	Human perception	Linguistics Voice Face
[60]	Communication skills	Human perception	Linguistics Voice Face
[61]	Communication skills, interpersonal skills, leadership, persuasion, negotiating, personality, overall hiring recommendation	Human perception	Linguistics Voice Face
[62]	Speech fluency	Human perception	Linguistics Voice
[63]	Impression management strategies	Self-report	Face
[64]	15 competencies related to interpersonal skills, problem solving, and personality and work style	Human perception	Linguistics

This study extends IS research on automated human screening, using design science research [DSR; 20,21] similar to that which has been successfully used in that domain. DSR is an appropriate approach because this study investigates a potential socio-technical solution—in this case, for assessing signals of job-performance at the application phase in a cost-effective and objective manner. Whether *objective* indicators of job performance can be efficiently identified in the job application process from AVIs is the critical question in our investigation.

The remainder of this paper will proceed as follows. First, we describe our DSR methodology in the context of assessing job candidates in more detail. We then review relevant literature on signalling and search costs in personnel selection to describe the problem domain and highlight the motivation for our research. We then follow with the meta-characteristics of a solution and describe the implementation. Following this, we instantiate a proof-of-concept prototype and evaluate its efficacy. We conclude by discussing the potential of this design and consider ethical implications of this approach.

2. Research methodology

The goals of this research included defining a novel problem space, formulating design principles grounded in theory, and demonstrating

these principles through a proof-of-concept instantiation. We therefore chose to adopt a DSR approach to facilitate achievement of these goals. Scientific contributions in DSR are typically *descriptive* (e.g., framing the problem space) and *prescriptive* (e.g., generating a design solution) [20]. The generalizable knowledge contributions of DSR include 1) a useful description of the problem space, 2) a solution that leverages important properties of the problem space, and 3) evidence of the potential of the proposed solution. These knowledge contributions constitute what has been termed "design theory" [22]. Design theories are tested and refined over time as future research extends the problem space framing or the generated design to improve solutions or adapt the solution to similar contexts [21].

Following Peffers et al.'s [23] DSR framework, we describe the problem domain, define objectives, propose an artifact design to meet these objectives, and then instantiate a prototype. Our evaluation follows similar DSR research [e.g., 3,24,25] in that our purpose is to demonstrate the feasibility of the proposed solution. We frame the candidate assessment problem space in terms of personnel selection signalling, search costs, behavioral evaluation, and thin-slicing concepts. We draw from this foundation when proposing a design theory for assessing a substantial pool of candidates. This is summarized in a set of meta-characteristics that for convenience we term systematic talent recognition and measurement (STREAM).

Fig 1

3. Objectives of new job application screening systems

In the introduction, we explained how our research question is motivated by the organizational need to detect candidates' signals in a reliable, validated, and cost-effective manner during personnel selection. While prevailing studies have predominantly examined predictive capacities of automated analysis of video interviews, they have often overlooked design characteristics necessary to enhance assessment validity and diminish decision-making biases. Thus, we seek to establish a cohesive design framework for valid technology-based job application processing.

In this section, we elaborate on personnel selection from a signaling perspective and discuss its validity and costs to further identify the problem space.

3.1. Signaling in personnel selection interactions

Because job seekers and hiring organizations each aim to further their individual interests, their interactions tend to feature misaligned goals. This misalignment can deter both parties from sharing comprehensive and accurate information. From this perspective, personnel selection can be framed as an uncertainty reduction problem [26]. Each party tries to reduce their uncertainty about the other by obtaining and evaluating *signals.* For businesses hiring candidates, these signals range from claimed educational experience and professional certifications listed in résumés during initial screenings to the displayed social characteristics and organizational fit communicated in interviews after the candidate pool has been winnowed.

While organizations use these signals to infer candidates' unobservable qualities, candidates also adapt their signals to match organizations' selection criteria [10,27]. This impression management (IM) can be both honest and deceptive [28]. During honest IM, candidates emphasize their positive attributes relevant to the job. In contrast, deceptive IM involves exaggerating, fabricating, or omitting facts, resulting in low signal reliability [28]. This in turn motivates organizations to constantly seek out costly qualities and hard-to-fake signals [10]. Costly qualities are those that require significant investment, such as obtaining advanced education or building a strong reputation so that references provide exemplary reports. However, since candidates can send signals that only *appear* costly (e.g., falsely claiming educational achievements), hiring organizations also want signals to be hard to fake. For instance, personality test results may be easy to fake, but signals of certain personality traits may be harder to fabricate in interviews. Similarly, high scores on cognitive ability tests can be hard to fake because candidates must possess the necessary cognitive prowess to perform well.

The new generation of job application–processing systems, then, needs to be selective when it comes to which new signals to use when evaluating candidates. Throwing every possible variable into a predictive model could lead to high initial predictive ability but low robustness to dishonest signals. Thus, we propose that the first objective for a new generation of AVI-based candidate evaluation systems is to focus on signals that are robust enough to withstand manipulation attempts.

Solution Objective 1. ***Capture Additional Robust Signals at the Application Stage***

3.2. Validity in personnel selection methods

A costly quality with a hard-to-fake signal is of little value if it is not valid—i.e., if it does not adequately predict future job performance. A wide variety of assessment techniques have been and will continue to be developed to assist with signal capture, but these techniques require validation to ensure that the signals they capture legitimately forecast a potential hire's job performance [29].

Although different jobs require different skills and characteristics, certain candidate evaluation methods tend to reliably predict future job performance across job types. Schmidt & Hunter [14] assessed the validity of candidate evaluation methods such as general mental ability (GMA) tests, work sample tests, conscientiousness tests, interviews (structured/unstructured), and peer ratings. Some of these measures focus on the use of a specific instrument (e.g., interviews, peer ratings), while others target specific constructs (e.g., GMA, conscientiousness). They ranked the top five most valid evaluation methods as 1) work sample tests, 2) cognitive ability assessments, 3) structured interviews, 4) peer ratings, and 5) job knowledge tests. Sackett et al. [15] revised

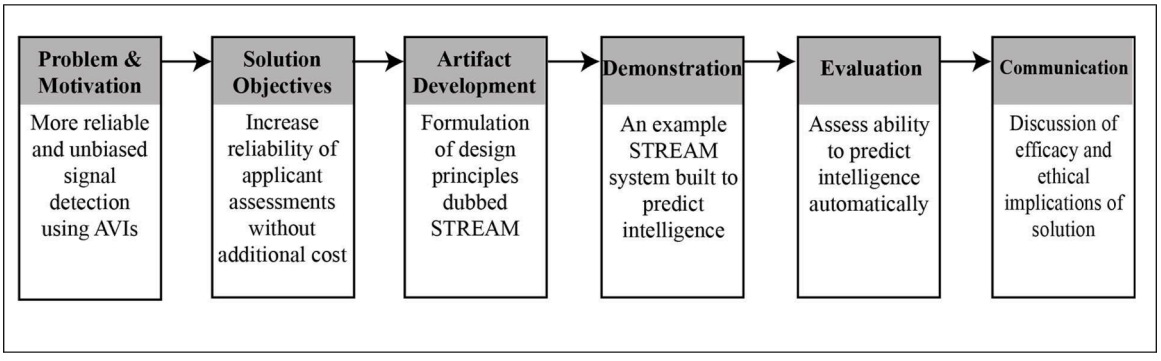


Fig. 1. DSR Process for STREAM (adapted from Peffers et al., 2007).

this list, accounting for systematic overcorrection in previous studies, and found that 1) structured interviews, 2) job knowledge tests, 3) empirically keyed biodata, 4) work sample tests, and 5) cognitive ability tests (e.g., GMA tests) and integrity tests comprised the most valid evaluation methods (cognitive ability and integrity tests tied for 5th).

In addition to these highly predictive evaluations, employers often assess attributes like personality, social skills, and interests [30]. Such evaluations help them gauge not just the alignment between the person and the job (person-job fit) but also the compatibility between the individual and the broader organizational culture (person-organization fit) [10]. However, these attributes tend to have comparatively low predictive validity [15].

When looking for valid signals, it is important to note that *structured* interviews top the list of valid methods for predicting job performance. Structured interviews are crafted to standardize the interview experience—every candidate undergoes the same sequence of interactions, questions, modalities, and scoring criteria [31]. Additionally, the questions asked are grounded in a job analysis aimed at identifying the most critical job-relevant traits. Structuring interviews not only minimizes hiring biases but also enhances the reliability and objectivity of interview outcomes [32].

For optimal evaluation reliability, a solution should exhibit the same consistency and validity found in structured interviews. Questions cannot all be exactly the same across industries or positions, because job-related experience is an important differentiator [15]—questions about job-specific skills should probably be asked even early on. However, questions, modalities, and scoring should be internally consistent for any given job if the evaluation is to produce the most valid and reliable signals.

Solution Objective 2. ***Use Valid and Standardized Evaluation Methods***

3.3. Search costs in personnel selection

Valid evaluation methods are important for a new generation of job application systems, but manageable search costs may be equally important if more top candidates are to be identified at the earliest stages. *Search costs* encompass all of the resources expended to gather essential information for decision-making [33]. In hiring, both recruitment and selection incur search costs. Recruitment involves strategically promoting an organization and its roles to attract high-quality candidates, while selection focuses on the evaluation of each candidate to identify the most qualified [34]. In this paper, we focus on search costs that arise from the selection process, including manual review of applications, signal validation, and assessment tools.

Larger pools of applications demand greater review cost, or fewer resources committed to reviewing each application. Overly limited resources prompt application reviewers to apply potentially biased heuristics, which can disenfranchise certain groups [35].

The quest for hard-to-fake signals further increases search costs. To discourage and detect candidates who send deceptive signals, organizations incur costs to perform background checks and contact references. They protect the validity of skills exams through proctoring and related efforts to combat cheating [36]. Even these costly efforts may fail to deter deception [37].

Each assessment tool used to gather signals beyond the traditional application review incurs additional cost to develop (or adopt), deploy, and maintain [38]. These costs are the primary reason why candidate pools are first winnowed using relatively low-cost methods, such as résumé and application review. Since each additional assessment tool incurs an expense, hiring organizations must weigh the costs against the marginal gains of obtaining this additional information.

Additional assessments at the application stage can also dissuade applicants. Candidate perceptions of the selection procedures can increase search costs by affecting candidate perceptions of job attractiveness [39]. Additional assessments can increase candidate opt-outs and attrition because candidates perceive the process to be overly time-consuming and tedious [11].

The advent of internet-centered application processes and computerized testing has helped reduce some of these search costs. Computerized testing reduces search costs by automating assessment administration and allowing assessments to be deployed earlier in the process to a larger group of candidates. Computerized testing does not eliminate search costs because assessments still must be developed internally or purchased from third-party vendors, and cheating must still be deterred. But inasmuch as concerns about cheating can be minimized, a computerized work sample test, a cognitive ability test, or a similar test is a scalable and somewhat cost-effective method of assessing a single candidate trait.

To assess several factors such as GMA, job skills, and personality, separate instruments would need to be used, which increases search costs to untenable levels, especially in terms of applicant attrition. A holistic solution should be able to assess a range of characteristics at low cost, using a single assessment. These characteristics should be ascertained using a thorough job analysis where employers identify the knowledge, skills, and abilities required to successfully complete duties [40].

In summary, while extant research emphasizes the validity of existing candidate assessment methods, it neglects the costs tied to implementation. For example, in-person structured interviews are efficacious but are limited in their application to only a fraction of candidates because of their resource-intensive nature. An optimal solution should accurately forecast future job performance and handle a large volume of candidates without a substantial cost increase. The ideal approach should reduce hands-on involvement, both in administering the assessment and in assessing results.

Solution Objective 3. ***Ensure Holistic Cost Efficiency at Scale***

In the next section, we prescribe a socio-technical framework consisting of specific meta-characteristics for a new generation of job application systems that meet these objectives. We propose that systems that implement this framework will improve organizational ability to identify the best candidates at the application stage.

4. Artifact development

Although artifact development is just one facet of DSR, crafting and scrutinizing such an artifact to address specific challenges is central to this approach [41]. In this section, we champion AVIs as tools adept at meeting the objectives listed previously. We detail foundational attributes for AVIs and demonstrate their implementation using a proof-of-concept example. The initial three principles are informed by current automated AVI studies, while the subsequent attributes are inspired by theories in personnel psychology and communication. Collectively, these insights shape a guiding framework for the future development of automated AVI systems.

4.1. Asynchronous video interviews

A primary component of our solution is the ability to assess comprehensive signals early in the selection process while minimizing costs. In response to this, we elected to use an existing technology, AVI, as the foundation of our artifact. Although the specific design may vary [2], the premise is the same for all AVI platforms—job candidates respond to computer-delivered interview questions while being recorded by the camera on their computer or mobile devices; there is no interaction between candidates and interviewers; responses are generally short (e.g., several minutes); and recorded responses are uploaded to a central platform where they can be reviewed by hiring managers.

The traditional value proposition for AVIs is in efficiency gains—employers can interview more candidates earlier in the hiring

process, can review recorded responses at their convenience, and can skip interview responses of candidates that do not appeal to them [42]. In addition, AVIs, which encompass both verbal and nonverbal cues, provide a richer sense of applicant social presence than other forms of asynchronous communication [43], while also discouraging deceptive impression management strategies [44]. This enhanced depth and authenticity may make AVIs a more effective tool for predicting future job performance than written assessments [45]. These advantages prompted us to implement a proof-of-concept AVI system that piggybacks on existing assessment procedures to maximize efficiency.

- Meta-characteristic 1: Utilize assessment methods that maximize efficiency through scalability and robust data collection.

4.2. Automated behavioral analysis

The idea that intangible qualities can be assessed from short video recordings is informed by research on *thin slicing*. A meta-analysis by Ambady & Rosenthal [46] showed that after observing thin slices of behaviors shorter than 5 min, humans make judgments about clinical and social outcomes that are quite accurate, with an effect size of 0.39. Although thin-slicing research is based on human judgments, results indicate that verbal and nonverbal behaviors of targets do signal objective measures of job performance. For example, sales effectiveness [47], competence [48], and intelligence [49] have been assessed at above-chance levels when observing thin slices of behaviors. Although above-chance judgments have limited practical value, this does suggest that even a minimal manual review of AVI-generated videos could yield additional insights.

Further, thin-slice judgments tend to be intuitive, and accuracy can be impaired by various factors, such as sadness [50] and tasks that involve reasoning [51]. Additionally, while AVI responses may be brief, manually reviewing a substantial number early in the hiring process can incur significant search costs.

In response to inefficiencies concerning manual review of AVI recordings, technologists have responded by automating candidate evaluation through the use of behavioral analysis. For instance, studies have used AVI responses to predict perceptions of personality [52,53] and communication skills [53]. Table 1 outlines this research on automated AVI systems.

Most automated interviewing research assesses three channels: voice dynamics, language characteristics, and face/head movements. Features such as eye and body behavior can also be automatically sensed. However, these features are more challenging to reliably capture, since they are easily obscured by eyeglasses or camera angles.

Computer vision technologies can process images of the face and interpret facial emotions. Many automated face analysis tools have become available to researchers and practitioners. Face behavior analysis is now readily applied in marketing [65], distracted driver awareness [66], deception detection [5], and many other human-computer applications.

The voice is strongly associated with affect [67] and is commonly used as a signal to interpret human behavior and emotions. Early research on voice generally relied on manual coding and focused on basic features such as pitch and loudness. Software applications like Praat [68] and OpenSmile [69] now allow extensive feature extraction in near real-time. The features derived using applications like these have been used to analyze and interpret a wide variety of human behaviors. For instance, Braga et al. [70] used Praat and machine learning algorithms to successfully detect Parkinson's disease from voice recordings. Other researchers have studied cognitive load recognition [71] and likability recognition [72] from speech.

Linguistic analysis tools such as Linguistic Inquiry and Word Count [LIWC, 73] and Structured Programming for Linguistic Cue Extraction [SPLICE, 74] can also provide insight into what is said. Although the linguistic analysis does not necessarily provide an understanding of the semantics of spoken communication, it can be used to detect meta-characteristics such as speech complexity, word count, emotions, and various other psychological dimensions. Features extracted from tools like LIWC and SPLICE have been found to signal cognitive and affect mechanisms [75] and used to predict personality traits [76], application success [77], and deception [78], among other outcomes.

We therefore advocate for the use of automated and structured behavioral analysis to assess candidates, which eliminates costs associated with manual assessment of interview responses. Our instantiation of STREAM is equipped with feature extraction tools and decision support algorithms that support the automated evaluation of candidates.

To generate linguistic measures, verbal responses are first transcribed. Since automated speech recognition can produce errors that affect language measure derivation [79], we opted to use human transcriptionists for this initial proof of concept to maximize data quality. After the transcription has occurred, the text is processed using LIWC and SPLICE to extract language measures. LIWC provides measures of psychological dimensions, and SPLICE provides measures of language complexity. In total, 229 linguistic features are generated, which include psychological dimensions, language complexity measures, and meta-statistics (e.g., word count).

To extract vocalic data, audio loudness is first normalized so that the perceived loudness over the duration of each recording is consistent. This offsets loudness variation caused by different microphone settings. Voice features are then extracted using OpenSmile [69]. These include theory driven features outlined below as well as more nuanced features commonly used in speech recognition, such as line spectral pairs and mel-frequency cepstral coefficients (MFCC). OpenSmile aggregates each measure over the full response to summary dimensions, including average, variance, min, and max. We also calculate the speech rate of every response by dividing the total word count by the response duration. In total, 2284 voice measures are derived. The current instantiation did not include face analysis due to environmental characteristics that affect tracking accuracy (e.g., inconsistent lighting, camera angle, camera quality).

- Meta-characteristic 2: Reduce manual intervention to increase efficiency.

4.3. Response verification

The cost to administer standardized assessments has been significantly reduced through unproctored web-based testing. This approach allows a variety of assessments to be deployed to more candidates earlier in the hiring process. However, the unproctored nature of web-based testing that facilitates scalability also hampers the reliability of the assessments. Although the degree of cheating is difficult to accurately estimate, Steger et al.'s [80] meta-analysis of proctored versus unproctored exams found that scores are generally higher for unproctored settings. The study also found that score differences between proctored and unproctored exams increase further when answers are easy to obtain using the internet. Online proctoring and validation testing are options to ensure the reliability of assessments, but these raise the cost to evaluate candidates. We therefore propose that to reduce search costs, candidate assessment should ensure verifiable identity and actions. In addition to the video that is recorded during each interview, our instantiation restricts the preparation time for each question. This approach ensures that the candidate's response stems from their own capabilities rather than being influenced by external coaching.

- Meta-characteristic 3: Assessment of response credibility should be verifiable and low cost.

4.4. Predictors of job performance

So far, we have advocated for the use of automated AVI assessments

and delineated the design principles that support this stance. The meta-characteristics have largely summarized the literature in Table 1 that uses automated behavioral analysis to assess AVI responses and have emphasized the potential benefits of this emerging technology. Our design framework further contributes to the literature by underscoring the importance of what is being assessed and how models are trained. Several important trends emerge when evaluating current automated AVI research.

To connect behaviors extracted from AVI recordings to outcome measures, much of the current research relies on manually generated outcome data produced by third-party raters. Although thin-slicing research indicates that humans can make fairly accurate assessments based on short recordings, within the personnel selection environment, organizations must be particularly careful when relying on machine-driven decision-making that uses human-derived training data [81]. Human-induced bias in machine learning models has been documented in many industries and remains problematic [82]. Organizations are potentially increasing their legal liability by using human-generated training data when hiring.

Further, many studies train models to predict "hireability" or general competence scores. This not only risks introducing human bias but also restricts the technology's generalizability. One characteristic of structured interviews that makes them such a valid selection instrument is customized questions based on the specific needs of a job [32]. Because each job requires a unique set of knowledge, skills, and abilities, basing the performance of a selection algorithm on generic constructs, such as hireability, limits the deployment of the technology. To increase the generalizability of automated AVI systems, models should be developed to assess well-defined constructs (i.e., signals). Organizations should then determine how much weight to assign to the various construct models based on a thorough job analysis. In this way, interviews are customized to assess candidates for a variety of jobs based on the needs of each position.

Finally, the constructs listed in Table 1, such as personality, communication skills, and leadership, are all valuable signals of employee success and have varying degrees of importance depending on the requirements of each job. However, the Table 1 constructs are not necessarily those found to be most predictive of future job performance. For reference, conscientiousness is the personality trait that has been found to most reliably predict job performance, and GMA is found to be 48 % more important than overall conscientiousness [15]. Based on a thorough job analysis, evaluation of candidates should initially focus on constructs that are historically shown to be most predictive of job performance. However, general trends should not override idiosyncratic differences. Over time, industry- and job-specific contexts should inform the signal selection of a STREAM-based system. For instance, conscientiousness may prove to be more predictive of performance than GMA or other factors for particular job types.

Thus while current research provides a foundation, STREAM systems must initially identify and iteratively refine the signals that are most relevant and reliable predictors of job performance.

Fig. 2 outlines our proposed approach, where signal evaluation models (e.g., communication ability, personality, intelligence, task knowledge, etc.) are first trained. Organizations then determine which models to use and their associated weight based on an analysis of job duties.

For this proof of concept, we demonstrate how a signal evaluation model can be created in a way that maximizes objectivity while minimizing human-induced bias. For this, we chose to focus on a quality that is valid yet not specific to any one job. GMA has been found to be an important predictor of job performance [14,15,83–86]. It predicts both short-term and long-term job success and has even been found to be more predictive of job performance than job experience or quantitative, verbal, and technical aptitudes [85]. [85] theorize that an increased ability to "acquire more job knowledge and acquire it faster" (p. 170) enhances job performance of higher GMA workers.

Sometimes, *intelligence* refers to innate potential and *GMA* to abilities developed over time [87]. We nevertheless use these terms interchangeably in this study, adopting Gottfredson's [88] definition of intelligence: "*a very general mental capability* that, among other things, involves the ability to reason, plan, solve problems, think abstractly, comprehend complex ideas, learn quickly and learn from experience" (p.13, emphasis added).

GMA testing has been conducted extensively since the early 1900s, but the proliferation of these tests stems from the U.S. Army's large-scale assessment efforts during World War I [89] and has iteratively improved since then. GMA is traditionally assessed using standardized paper-and-pencil or computerized testing [90]. It can also be indirectly

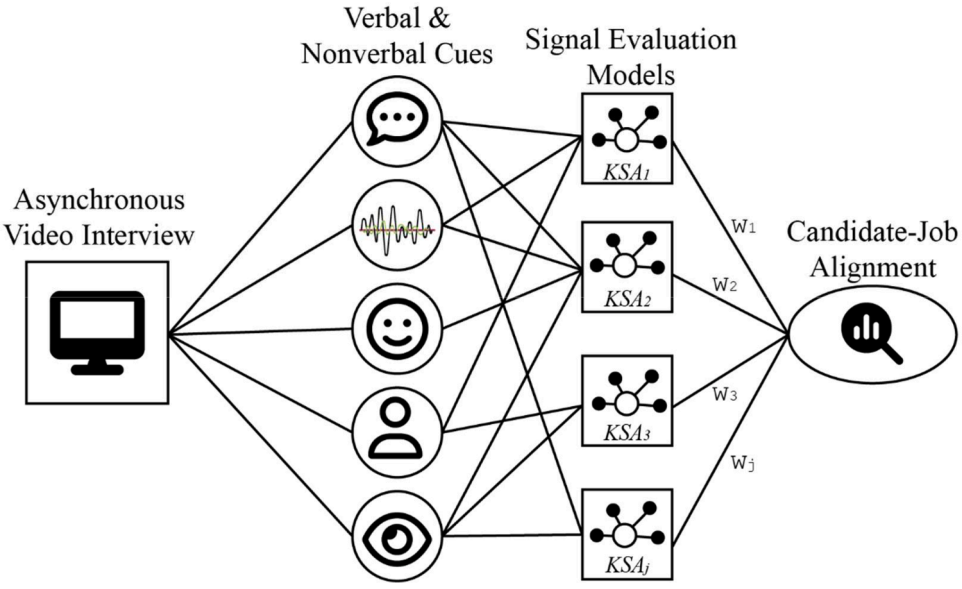


Fig. 2. Proposed two-stage fusion methodology.
Note: The diagram represents the process of extracting verbal and nonverbal cues from AVI recordings and then using these cues as input into separate models to automatically assess job-relevant knowledge, skills, and abilities (KSA). The Signals Evaluation Models implemented and their weighted values will vary based on job-analyses performed by organizations.

assessed using situational judgment tests, assessment centers, and interviews. For instance, standardized assessment scores of GMA are positively correlated with interview performance evaluations [91]. This indicates that interviews might be used as orally administered cognitive ability tests [92]. Huffcutt et al. [91] propose four reasons that interviewer ratings correlate with cognitive ability: 1) intelligence stands out and is remembered by the interviewer; 2) interviewees with a higher intelligence may be better able to select more appropriate self-presentation strategies; 3) these interviewees are able to answer questions more effectively; and 4) more intelligent people are likely to have stronger work histories and educational background presented in their application materials, which may bias interviewer ratings. For the current proof of concept, we elected to use an established GMA assessment instrument as the basis for training prediction models.

- *Meta-characteristic 4: Target traits that are most predictive of job performance.*
- *Meta-characteristic 5: Limit the use of subjective human judgments when training models.*

4.5. Selecting and evaluating behavioral cues

Theories from the fields of communication and psychology suggest that both verbal and nonverbal behaviors can provide insights into the underlying traits of the message sender. For instance, a combination of body movements and vocal patterns can suggest dominance [93]. Similarly, when a person is under increased cognitive load, their speech becomes more tense [71]. When assessing characteristics based on behavioral cues, it is crucial to focus on the most relevant indicators.

In the development of our prototype, the selection of specific linguistic and vocalic signals is based on prior findings related to GMA. According to the PASS theory of intelligence, human cognition consists of three functional units [94]. The first one is attention-arousal, which helps maintain alertness and avoid distraction [95]. Individuals with high GMA tend to focus more cognitive resources on the interview. Therefore, we expect GMA to negatively correlate with disfluencies and the use of filler words and positively correlate with the response duration [96], speech rate [97], and the use of words that demonstrate cognitive processes. The second functional unit is in charge of simultaneous and successive mental processing [95]. Individuals with high GMA can better manage sequences, such as words and sentences, so we expect GMA to positively correlate with lexical diversity and the use of complex words and sentence structures.

The third functional unit is planning, which helps individuals choose strategies for problem-solving [95]. High GMA enables more effective planning, so interviewees adopt better self-presentation strategies to manage their image and information. Based on uncertainty reduction theory [98], individuals seek to reduce the uncertainty and increase the predictability of their communication partners' behaviors. In an ideal response, speakers are confident in the accuracy of their information [99], so certainty is a desired verbal feature that high-GMA individuals seek. Because adjectives and adverbs provide additional description of the nouns and verbs that they describe and help reduce uncertainty, we also expect that high-GMA individuals utilize adjectives and adverbs for impression management, which leads to a more expressive language style. In the vocalic channel, high pitch variability is associated with favorable perceptions, such as benevolence and competence, while high amplitude variability is related to perceived vocal hoarseness and low voice quality [100]. Therefore, we expect high-GMA individuals to display high pitch variability and low amplitude variability to obtain a positive self-image in the interview. Behavioral cues of GMA and the expected relationship (positive/negative) with GMA are outlined in Table 2. These signals should be prioritized during artifact development.

- Meta-characteristic 6: Prioritize theoretically grounded behavioral cues based on the construct being evaluated.

Table 2
Summary of Expected Behavioral Signals of Intelligence.

Behavioral Signal	Expected Relationship with GMA
Speech hesitations and errors (filler language)	Negative
Response duration	Positive
Speech rate	Positive
Cognitive processes	Positive
Lexical diversity	Positive
Complex language (word length & SMOG score)	Positive
Hedge and uncertainty	Negative
Pitch variability	Positive
Amplitude variability	Negative

4.6. Interview question types

The tempting choice for implementing AVIs may be to simply re-use available easily accessible or preferred questions, but this path risks not eliciting diagnostic behavioral cues. Literature in personnel selection, communication, and psychology reveal the diverse set of traits that can be obtained through interviews, which demonstrates that a variety of job-fit criteria can be evaluated during a single AVI interview. However, the types of questions asked during interviews are important determinants of the information that can be obtained from interviewees [32]. Therefore, to maximize the reliability of signals captured by AVIs, question sets should be specified for constructs of interest. For instance, when attempting to predict the personality trait of conscientiousness, interview questions should focus on goal-directed and detail-oriented behaviors [32].

Exhaustively considering all possible types of questions would be optimal, but the complexity involved makes it cost-prohibitive. But to ensure validity, question sets should at least be linked to the desired trait. For instance, the most widely studied employment interview question types include past-behaviour questions (PBQs) and situational questions (SQs). PBQs ask interviewees to describe situations from their past (e.g., "Tell us about a time when…"), and SQs ask interviewees to address a hypothetical scenario (e.g., "What would you do if…"). For both question types, the assumption is that responses will help to predict future actions. PBQ and SQ types are both found to correlate with cognitive ability, although the degree of correlation between the question types and cognitive ability may vary based on the type of job [91, 101]. PBQs also correlate more with job experience compared to SQs, but SQs are more related to job knowledge [102].

Existing research that attempts to automate AVI evaluation places varying degrees of importance on the use of specific question types. Suen et al. [63] use past-behavior questions to detect impression management tactics. Chen [52] also uses specific past-behavior questions to assess communication skills, interpersonal skills, leadership, and persuasion. Suen et al. [53] used five communication-specific questions to automatically predict communication skills but also used the same questions when attempting to predict personality traits. Niemitz et al. [103] evaluate perceptions of AVIs based on different question delivery formats (text versus video) but do not consider the types of questions asked. Other research [e.g., 58,60,104] does not discuss the theoretical value of selecting specific questions when attempting to elicit a construct of interest.

Because of the importance of question selection and the mixed attention placed on this, we propose that question selection be a critical component of a STREAM-compliant design.

To exemplify this, our instantiation of a STREAM system outlined in the next section relies on a five-question panel (see Table 3) to elicit signals of GMA. The opening question (*Describe your greatest strength.*) is designed as an orienting question to get users used to the system. This question is not used for analysis. The remaining questions are divided between PBQs and SQs. Each question type has previously been found to correlate with GMA [32]. Additionally, the PBQs and SQs selected for the current instantiation focus on problem-solving tasks intended to

Table 3
Interview Questions.

Question Type	Interview Questions
General	Describe your greatest strength.
PBQ	Describe a situation that required a number of things to be done at the same time. How did you handle it? What was the result?
PBQ	Tell me about a time when you came up with a new or innovative way to solve a problem.
SQ	What would you do if your manager asked you to perform a task that you'd never done before?
SQ	How would you handle a situation in which you had a conflict with a coworker?

elicit GMA. The ability to include additional questions in the interview protocol highlights the efficiency of AVIs for assessing multiple characteristics. Along with the questions listed in Table 3, which are designed to elicit GMA, hiring managers could also include questions related to personality, sociability, organizational fit, or any other critical constructs identified through a job analysis.

- Meta-characteristic 7: Include question types that are theoretically likely to indicate the selected intangible qualities.

5. System instantiation summary

To summarize, the proposed artifact consists of seven meta-characteristics intended to increase the efficiency, validity, and fairness of selection while limiting search costs. The first three characteristics establish support for our argument that AVIs are a viable technology for robust and scalable candidate evaluation. The remaining principles draw from theory in personnel psychology and communication to establish a framework for the evolution of AVI development. These characteristics focus on the first stage of fusion, as shown in Fig. 2, which encompasses interview delivery, response capture, cue extraction, and signal model training. Table 4 summarizes these characteristics and indicates how we have implemented them in a proof-of-concept prototype.

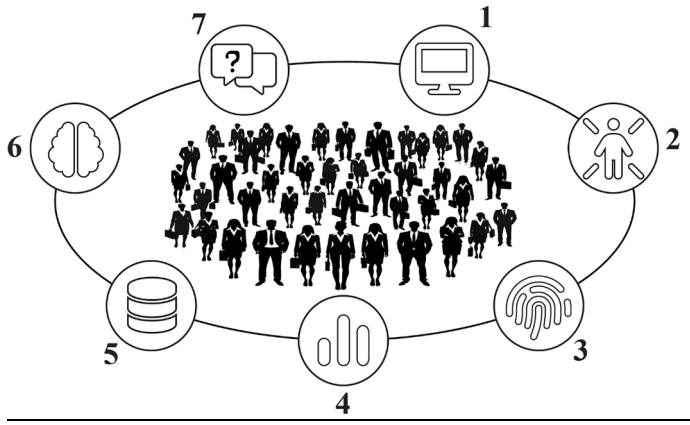
6. Artifact demonstration and evaluation

To examine the implementation of the prescribed meta-characteristics and the ability of the system to predict GMA, a mock interview was conducted. Study participants ($N = 121$) were recruited using Prolific.com, which is a web-based company that provides high-quality research participants. Participant selection included only those who were currently living in the United States, were fluent in English, and were willing to be video recorded. Participants responded to interview questions using their personal web cameras and in locations of their preference. This replicates the type of environment variables encountered by standard AVI platforms. Seventeen subjects were removed from analysis due to technical issues (e.g., corrupt video files) or not completing all parts of the experiment process. The final sample consisted of the remaining 104 participants, which included 52 males, 50 females, and 2 subjects who did not report their gender. Excluding one participant who did not report age, participants' average age was 37.74 ($SD = 12.89$; min = 18; max = 75). Participants were paid approximately $10 for their participation.

Analysis includes both regression and prediction tasks. The regression analysis is intended to evaluate the relationship between GMA and behavioral cues selected based on prior research. The prediction task is intended to assess the overall predictability of GMA based on an array of behavioral cues extracted from interview responses.

Table 4
Meta-characteristics and Implementation.

Systematic Talent REcognition and Measurement (STREAM) Process



Meta-Characteristics	Implementation
1. **Capture signals at scale.** Use assessment methods that maximize efficiency through scalability and robust data collection.	Base the system instantiation on AVIs since the technology is scalable and provides the ability for robust signal evaluation.
2. **Automate signal processing.** Reduce manual intervention to increase efficiency.	Increase the efficiency of candidate evaluation by automatically extracting behavioral cues from AVIs and using them to build classification models.
3. **Ensure signals are hard to fake.** Assessment of response credibility should be verifiable.	The proctored nature of AVIs ensures that the identity of the person being interviewed can be verified and responses to questions are not coached.
4. **Estimate reliable performance indicators.** Target traits that are most predictive of job performance based on analysis of job requirements.	Create a system that can predict general mental ability (GMA).
5. **Use unbiased and valid training data.** Limit the use of subjective human judgments when training models.	Use an established GMA assessment instrument as the basis of model training.
6. **Rely on signals that generalize.** Prioritize theoretically grounded behavioral cues based on the construct being evaluated.	Extract and evaluate verbal and nonverbal behaviors associated with GMA.
7. **Use questions that elicit useful signals.** Include question types that are theoretically likely to indicate the selected intangible qualities.	Use past-behavior and situational questions targeted toward problem-solving.

6.1. Procedure

The data collection for this study is conducted fully online. After completing a technology check and consenting to the study, participants complete a short pre-survey. They are then directed to a video which explains what they will do for the study. The video instructs participants to complete a short GMA assessment, which will then be followed by an AVI. The participants are also told that they will receive an additional payment of $5 if their performance on the assessment and interview is better than 75 % of the other participants' performance. This instruction is a slight deception intended to motivate performance. All participants ultimately receive the bonus payment.

Following the video instructions, participants are directed to the GMA assessment. After completing the assessment, they watch a second video that explains how to use the interview system. They then respond to the five interview questions outlined in Table 3. The interview system mirrors commercial asynchronous interview systems—after an interview question is revealed, participants have 30 s to consider their response and 60 s to respond to the question. The system will stop recording after 60 s. If participants finish a question before 60 s, they have the option to proceed to the next question. Following the video interview, participants complete a short post-survey that asks about their experience with the study, including their motivation to perform. When asked in the post-survey if the $5 bonus motivated their performance, 75 % of participants responded "Agree" (25 %) or "Strongly Agree" (50 %).

6.2. GMA assessment

The Wonderlic Personnel Test—Quicktest (WPT-Q) is used to gauge GMA. The full Wonderlic Personnel Test (WPT) [105] is commonly used in personnel selection to assess GMA [85]. The WPT-Q is found to reliably replicate the full WPT but can be administered online in a shorter amount of time [106]. The WPT-Q includes 30 questions, which must be completed in 8 min or less. Scores can range from 0 to 38. In the current study, WPT-Q scores range from 17 to 36, with an average of 27.73 and a median of 28 ($SD = 4.61$).

7. Analysis and results

7.1. Regression analysis

To examine whether verbal and nonverbal behaviors signal GMA, we first conduct regression analysis with behavioral variables selected based on theories and the existing literature. Individual models are constructed for each signal of interest. The regression model is specified as follows. $Behavior_{i,j}$ is the verbal or vocal feature (e.g., disfluency ratio) extracted from the video clip of the response by participant i to question j. $Score_i$ is the WPT-Q score achieved by participant i. Age_i and $Gender_i$ are participant i's age and gender, respectively. The missing demographic values are imputed with the average of other participants' values. X_j is a vector of dummy variables for question j. μ_i is the participant-level random effect, which follows a normal distribution. $\varepsilon_{i,j}$ is the error term.

$$Behavior_{i,j} = \beta_0 + \mu_i + \beta_1 Score_i + \beta_2 Age_i + \beta_3 Gender_i + \beta_X X_j + \varepsilon_{i,j}$$

$$u_i \sim N\left(0, \sigma_\mu^2\right)$$

$$\varepsilon_{i,j} \sim N\left(0, \sigma_e^2\right)$$

Regression results are reported in Table 5 and Table 6.

7.2. Regression results

The coefficients of the WPT-Q score are positive and statistically significant in the models of response duration, speech rate, and average word length, indicating that individuals with higher GMA have a longer response duration, a higher speech rate, and a higher average word length. Because responses are limited to 1 min, we also run the model of response duration with samples shorter than the time limit, and the relationship remains robust. These results are consistent with our expected directions. A long response duration and a high speech rate are related to the abundant cognitive resources allocated by the attention-arousal functional unit, and the use of long words can be attributed to better mental processing of sequences.

The coefficients of the WPT-Q score are also significant in the models of hedge and uncertainty and expressivity. However, the directions are opposite of what we expected. The higher GMA an individual had, the more hedge and uncertainty and the less expressivity the responses contained.

Because there are 12 regression models, we set the familywise error rate to be 0.1 and conduct Bonferroni correction, which is a conservative way to control Type I errors [107]. The results for response duration, hedge and uncertainty, and expressivity remain significant after Bonferroni correction. The next section on the predictive analysis will further demonstrate the usefulness of interview behaviors for assessing GMA.

7.3. Prediction analysis

Identifying behavioral indicators of GMA with regression analysis lays the foundation for predicting GMA with interview responses. We first formulate GMA prediction as a classification task. GMA measures (i.e., WPT-Q scores) are discretized into High and Low classes based on their median values. Participants (49.04 %) who have scored higher than the median are labeled as High in GMA, and the remainder are labeled as Low.

For this prediction task, we expanded the feature set to include language and voice signals beyond what was evaluated with regression. As described previously, 229 linguistic and 2284 vocalic features were derived from each response. To reduce the dimensionality of the dataset

Table 5
Regression Results.

	Dependent variable:					
	Disfluency ratio (1)	Filler words (2)	Response duration in second (3)	Speech rate (4)	Cognitive processes (5)	Lexical diversity (6)
WPTQ-Score	−0.001	−0.0005	1.001***	0.022**	0.022	−0.002
	(0.001)	(0.0005)	(0.247)	(0.008)	(0.049)	(0.001)
Age	0.0004**	0.0003*	−0.027	0.003	−0.017	0.0001
	(0.0002)	(0.0002)	(0.089)	(0.003)	(0.018)	(0.0005)
Gender (Male = 1)	0.011**	0.010**	1.780	−0.002	−0.917**	−0.004
	(0.005)	(0.004)	(2.243)	(0.076)	(0.450)	(0.013)
Marginal R Squared	0.098	0.090	0.160	0.076	0.488	0.016
Conditional R Squared	0.631	0.624	0.758	0.700	0.491	0.515

Note: *$p < 0.1$; **$p < 0.05$; ***$p < 0.01$.

Table 6
Regression Results Continued.

	Dependent variable:					
	Average word length (1)	SMOG (2)	Hedge and uncertainty (3)	Expressivity (4)	Pitch variability (5)	Amplitude variability (6)
WPTQ-Score	0.010**	0.050	0.672***	−0.003***	−0.091	0.0003
	(0.004)	(0.038)	(0.162)	(0.001)	(0.555)	(0.001)
Age	−0.002	−0.025*	0.008	−0.0001	0.225	−0.0005
	(0.001)	(0.014)	(0.058)	(0.0003)	(0.200)	(0.0005)
Gender (Male = 1)	0.024	−0.267	0.281	−0.007	6.378	−0.034***
	(0.037)	(0.347)	(1.474)	(0.007)	(5.043)	(0.012)
Marginal R Squared	0.120	0.031	0.158	0.191	0.031	0.076
Conditional R Squared	0.395	0.261	0.560	0.564	0.957	0.885

Note: $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

and reduce overfitting, we removed some less interpretable vocalic features, such as complex summary statistics of the linear approximation of the contour, interquartile ranges, outliers within the recordings, and the shape of the distribution. We further removed features that have a very small magnitude (absolute value lower than 1e-3) for more than half of the recordings. Some of these features correspond to sparse linguistic features related to specific topics such as religion or health. Others generally correspond to vocalic features with values of a very small magnitude. The standard deviations of these features (scaled by the mean absolute values) tend to be larger than those of the other features, suggesting that vocalic features with a very small magnitude can be easily varied because of environmental factors such as uncontrolled background noises and thus unreliable to make predictions about. We also removed features that correlate with another feature above 0.70 or below −0.70. After this stringent feature filter process, 74 vocalic features and 85 linguistic features remained. These features were then averaged across the four interview questions for each participant (the general interview question related to greatest strength was not included in the evaluation).

To predict high and low levels of GMA, 70 %, 15 %, and 15 % of the multimodal dataset were randomly split into training, validation, and test sets 50 times. For each split, we applied ANOVA F-test feature selection to the training set for further dimension reduction. ANOVA F-test compares whether there is a statistically significant difference in the means of two or more groups [108]. A higher F statistic indicates a higher ratio of between-group variability to within-group variability, hence a more discriminative feature, as the size of groups remains the same. Then, we selected the top K features with the highest F statistics in the training set and used these features in the validation and test sets for hyperparameter tuning and model evaluation. This Monte Carlo cross validation [109] data-splitting strategy allowed us to evaluate model performance more robustly.

We trained five popular machine learning classifiers: logistic regression (LR), support vector machine (SVM), random forest (RF), adaptive boosting (AB), and bagging (BAG) classifiers (the last being ensemble classifiers) with LR models as base estimators. Besides the number of selected features, each classifier also has its own set of hyperparameters that can affect model performances. We performed a grid search of hyperparameters by evaluating the model performances on the validation sets. Specifically, we averaged the accuracy scores on the 50 validation sets and select the optimal set of hyperparameters that achieves the highest accuracy score.

Additionally, we constructed a baseline classification score imitating random guesses. Specifically, we made random binary predictions for each test set using a Bernoulli distribution with the probability equal to the percentage of high labels in their corresponding training set. The average accuracy, F1 score, and AUC of the 50 test sets were our baseline.

Table 7 shows that GMA is reliably predicted with an average accuracy of 0.694 across all the five classifiers. LR achieves the highest accuracy of 0.704, significantly outperforming the random guess baseline of 0.489. Moreover, F1 scores range from 0.660 to 0.689, indicating a balanced performance of precision and recall. AUC of all the models is above 0.7. The performance results demonstrate the viability of the system and that it is possible to predict objective measures of job performance well above chance from recorded interview responses.

Table 7
Performance of GMA Classification.

Performance – Multimodal Model			
Classifier	Accuracy	F1	AUC
LR	0.704	0.688	0.759
SVM	0.692	0.660	0.757
RF	0.696	0.689	0.756
AB	0.697	0.689	0.761
BAG	0.679	0.660	0.741
Random Classification Baseline	0.489	0.500	0.500

Note: The highest performance metrics are in bold.

We compared the predictive power of the vocalic and linguistic modalities using the same method described previously and reported the analysis and results in the Appendix. The linguistic features were found to be more predictive of GMA than the vocalic features. We further provided preliminary evidence in the Appendix showing that PBQs are more telling of GMA than SQs.

To promote equity in hiring decision making, we examined whether the system discriminates against historically unprivileged groups, namely females and older individuals. The results in the Appendix indicate that our system produces fair predictions with respect to gender, age and ethnicity and complies with the 80 % rule set by the US Equal Employment Opportunity Commission (EEOC, 1979).

In addition to classification, we also formulated GMA prediction as a regression task and followed similar steps to train, validate, and test various regression models. We provided more details of the regression procedure and reported the average Pearson correlation coefficient (R), root mean squared error (RMSE), and mean absolute error (MAE) across the test sets in the Appendix. The results corroborate the classification performances and again showcase the system's viability.

7.4. Candidate ranking evaluation

Managers may also be interested in predicting the top-ranked candidates (i.e., the top 10 candidates from the interview pool). We evaluated the viability of this prediction task by ranking the candidates in a test set based on their probability of having a high GMA label output by the classifiers. We then compared this ranking against a ranking of the actual GMA scores.

Where T denotes the number of top candidates, two relevant metrics, recall at T and precisions at T, are used to evaluate the feasibility of this task. A higher recall indicates the system is less likely to miss out on good candidates, and a higher precision implies the system can help the organization focus its resources on better candidates. These metrics are

defined as follows.

$$\text{Recall at } T = \frac{\#\text{ of predicted top } T \text{ candidates that are actually in top } T}{\#\text{ of top } T \text{ candidates (including ties)}}$$

$$\text{Precision at } T = \frac{\#\text{ of predicted top } T \text{ candidates that are actually in top } T}{T}$$

In real life, T depends on the number of candidates that the organization would like to advance to the next round. In our evaluation, we attempt to select the top 50 % of candidates from the test set. We conduct simulations of random guesses to determine the baseline. Table 8 shows that the precision and recall metrics are all well above the random guess baselines, indicating that it is feasible to identify the top candidates more precisely using our system design. However, as highlighted in Appendix E, increasing the selectivity (i.e., reducing T) does lead to a decrease in model precision (e.g., from 0.735 at 50 % selection ratio to 0.600 for $T = 5$ and 0.533 for $T = 3$ for the AB classifier). While these values remain higher than random classification baselines, this trend should be considered a potential boundary condition for the STREAM system, as more selective criteria may impact overall accuracy.

7.5. Feature importance evaluation

To understand how each behavioral feature contributes to our predictions of GMA levels and rankings, we applied the ANOVA F-test to the full multimodal dataset ($N = 104$, 74 vocalic features and 85 linguistic features) and ranked the features using their F statistics as the importance metric. Fig. 2 visualizes the modalities and importance of the top 50 features; 24 vocalic and 26 linguistic features are among the top 50. While the quantities of the vocalic and linguistic features are almost equal, seven of the top 10 features are linguistic. These results demonstrate the strong predictive ability of the linguistic features compared to vocalic features.

Table 9 reports the top 10 features and the average value of these features for the high and low GMA groups. T-tests that compare the means of these features between the two groups are all significant, with p values lower than 0.001 for nine of the features and p values below 0.01 for the remaining feature.

Table 9 also reports the correlation between these features and the raw WPT-Q score. High GMA is associated with higher word complexity, more hedging, more words, less disfluency, and a greater number of plural nouns and conjunctions. Moreover, GMA is associated with several vocalic features involving the first-order differential of frequency-based measures.

Fig.3

8. Discussion

8.1. DSR knowledge contributions

The challenges and costs of the hiring process have long been a focal point in personnel selection literature. In response, various assessment tools, including automated AVI systems, have been developed. Viewing our work through the lens of DSR, we present a novel solution to this recognized problem. In such contexts, As Gregor and Hevner articulate, "Researchers must contend with a known application context for which useful solution artifacts either do not exist or are clearly suboptimal" ([20], p. 346). Throughout this paper, we have highlighted that our proposed STREAM framework establishes essential design aspects that current AVI research often overlooks. These include the predictive validity of the evaluated construct, the impartiality of training data, the choice of behavioral cues, and the significance of question types. The existing literature's gap in addressing these design principles results in less-than-ideal solutions. The future of AVI-based screening systems in research and practice needs established frameworks like STREAM to demonstrate validity and minimize bias.

Table 8
Performance of GMA Ranking.

	Performance—Multimodal Model	
	$T = 50$ % of test set	
Classifier	**Precision**	**Recall**
LR	0.733	0.685
SVM	0.715	0.667
RF	0.720	0.672
AB	**0.735**	**0.687**
BAG	0.702	0.656
Random Classification Baseline	0.57	0.533

Note: The best performance metrics are in bold.

Table 9
The Top 10 Features Selected From the Multimodal Dataset.

Ranking	Modality	Feature	Mean of high GMA	Mean of low GMA	Correlation with WPT-Q score
1	Linguistic	Number of words with 3 or more syllables	14.291	10.390	0.449***
2	Linguistic	Number of verbs that indicate hedging	1.368	0.750	0.359***
3	Vocalic	The 1st quartile of the 1st order differential of line spectral pair frequencies computed from the 3rd LPC coefficient	−0.014	−0.011	−0.348***
4	Linguistic	Word count	140.062	111.901	0.402***
5	Vocalic	The 2nd quartile of the 1st order differential of mel-frequency cepstral coefficients (MFCCs) 2	0.131	0.068	0.344***
6	Linguistic	Number of plural nouns	4.435	3.167	0.371***
7	Linguistic	Percentage of "uh"	0.009	0.021	−0.256**
8	Linguistic	Number of conjunctions	6.049	4.539	0.359***
9	Linguistic	Hedge and uncertainty (a combination of indefinites, demonstratives, hedging words, uncertainty words, and the Loughran and McDonald [2011] uncertainty and weak modal dictionaries)	21.015	15.865	0.387***
10	Vocalic	The 1st quartile of the 1st order differential of line spectral pair frequencies computed from the 2nd LPC coefficient	−0.018	−0.014	−0.271**

Note: The average value of these features for the high and low GMA groups and the correlation between these features and the raw WPT-Q score are calculated using the entire dataset ($N = 104$). *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

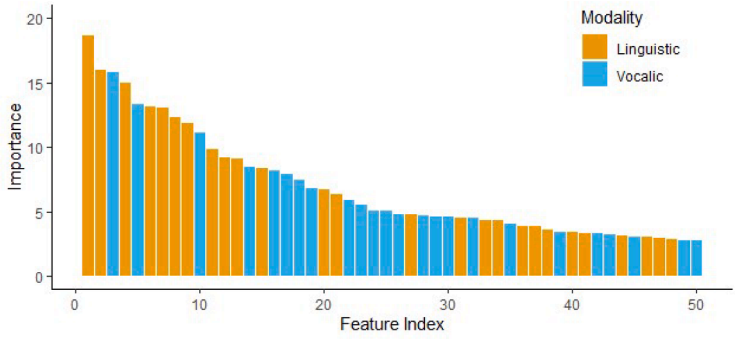


Fig. 3. Modalities Comparison of Top 50 Most Important Features.

Given that automated AVIs intersect with personnel psychology, verbal and non-verbal communication, and digital selection methodologies, our approach has integrated theories and findings from these domains. In doing so, we have aimed to define design principles firmly anchored in established theory. Our contributions align with what Gregor and Hevner categorize as a level 2 DSR contribution, emphasizing emergent design theory [20]. Beyond theory, we further our contribution by showcasing the practical implementation of these principles through the development and evaluation of a proof of concept.

Table 10 provides a summary of the DSR knowledge contributions of this.

During the evaluation of the prototype, GMA was predicted with AUC scores ranging from 0.74 to 0.76. Further, the candidate ranking evaluation provides evidence that candidates with the highest GMA scores can be identified beyond a binary high/low classification. From a DSR perspective, the results indicate the potential of the proposed solution by uniquely revealing that it is possible to evaluate GMA at levels significantly above chance using short video recordings.

However, our foremost aim in this research was to establish design principles as a foundation for shaping future automated AVI systems rather than to create a fully validated predictive model of job performance. It is important to note that we are not suggesting that GMA-specific AVIs be used to solely estimate job performance or that the presented instantiation is primed for immediate real-world application. The prototype's conception primarily served to explore the feasibility of implementing the prescribed design principles. While GMA is recognized as a reliable proxy for job performance, it remains just that—a proxy. To fully realize the system's potential for predicting job performance, more refinement is imperative, including the evaluation of other critical job performance indicators (e.g., job knowledge). Evaluation of our prototypical design, however, gives merit to our assertion that objective measures of job performance can be evaluated based on brief interview responses and without reliance on subjective training data.

Table 10
Knowledge Contributions of STREAM.

	Type of DSR Contribution (from Gregor & Hevner [20])	Description
1	Descriptive, unique framing of a problem space	Automated AVIs have been suggested as solutions for efficiently handling high-volume job applicant screening situations. We further contribute to this discussion by examining their impact on search costs and exploring the challenges associated with maintaining objectivity in their current implementations.
2	Prescriptive solution for the problem	Drawing from reference literature in personnel psychology, communication methodologies, and digital selection methodologies, the meta-characteristics of STREAM outlined in Table 4 are theoretically expected to improve the objectivity of automated AVI while also preserving efficiency and cost savings.
3	Prototypical instantiation and evidence for the concept	A prototypical instantiation demonstrated initial ability to objectively classify between high and low intelligence, which past research has identified as a key indicator of job performance.

8.2. Cues of GMA

The assessment of the prototype also sheds light on the significance of various behavioral characteristics, which provides direction for the development of subsequent AVI systems. Similar STREAM-based systems must identify cues that reliably reveal generalizable signals. In our proof-of-concept, theoretically selected language and voice cues were selected, and we found that response duration, speech rate, hedging/uncertainty, word length, and expressivity were significantly correlated with GMA test scores. We also conducted in-depth feature importance evaluation on a broader set of language and voice features not included in prior research. Our evaluation of the top 50 most important features aligns with extant literature in that GMA is communicated through a combination of verbal and nonverbal behaviors. However, the top 10 features were dominated by linguistic cues, suggesting that language may provide more insight than voice characteristics. [91] suggest that interviewees with higher cognitive ability can more appropriately select self-presentation strategies that influence interviewers. Since an interviewer was not present in the current study, the interviewees perhaps focused more on their spoken responses to questions and less on self-presentation strategies that would influence their voice characteristics.

Further, opposite to the expected relationship, hedging language was found to positively correlate with GMA test scores. Crompton [110] defines hedging as "an item of language which a speaker uses to explicitly qualify his/her lack of commitment to the truth of a proposition he/she utters." Our results revealed a positive correlation ($r = 0.359$, $p < 0.001$) between hedging verbs (e.g., assume, feel, seems, think) and GMA and a positive correlation ($r = 0.387$, $p < 0.001$) between GMA and a composite measure of hedging and uncertainty (e.g., maybe, might, could). These correlations could suggest that individuals high in GMA are uncertain of and/or not committed to their statements. However, through manual review, interview transcripts high in uncertainty and hedging language appear to reflect a more complex thought process. For example, one response to the SQ "What would you do if your manager asked you to perform a task that you'd never done before?" included the following statement (the response below is abbreviated to focus on hedging and uncertainty language):

"I might ask a few clarifying questions… depending on the answer… then I might put a few thoughts together … depending on the size of the project, I

might put together a project plan … I may do my own research … could look at other resources … could also engage a mentor."

This response is full of hedging and uncertainty language but reflects a thoughtful decision tree. Arguably, the participant lacks commitment to his or her propositions but only because the participant realizes that various routes can be taken as new information is gained (e.g., "depending on the answer," "depending on the size of the project"). In this case, the use of hedging and uncertainty language appears to reflect a broader understanding of the situational context and outlines possible steps to reach a successful outcome while insinuating that different paths to that outcome exist.

Alternatively, hedging can also be a useful communication strategy for harmonizing communication between sender and receiver [see, 111–113]. Given that high-GMA individuals are thought to excel in choosing impression management tactics [91], our study's participants might deliberately use hedging to influence the reception of their statements.

Future iterations of the system should explore the implications of sentence structure and other linguistic signals of thought complexity. This is especially important for cues where significance may not be immediately apparent. For instance, in the current study, plural nouns were among the top 10 indicators, yet the reason for this is not immediately clear. We speculate that the use of plural nouns might signal a more comprehensive grasp of a subject—for instance, discussing "algorithms" suggests a wider knowledge base than merely mentioning a single "algorithm." It could also imply a richer array of experiences. For example, someone speaks about "internships" rather than a singular "internship," which suggests they have engaged in multiple professional learning opportunities. It is important to note that the top features presented in Table 9 are based on our proof-of-concept study with a small online sample and low stakes, so future research should continue to explore these features and examine their generalizability.

8.3. Ethical implications and limitations

Though this study provided significant guidance in the development of automated AVIs for candidate screening, there are several important limitations that provide opportunities for future research to further validate and enhance the STREAM standard.

While algorithms are touted to neutralize human biases, they can inadvertently amplify them [114]. We acknowledge that despite our best efforts, our automated AVI design will likely not be entirely free from biases. However, the design principles were shaped to foster a more objective selection framework. Drawing on existing literature about interview design, we argue that the rigid structure inherent to AVIs serves as a means to enhance fairness in the selection process. Additionally, our theory-based meta-characteristics emphasize objectivity: targeting distinct knowledge, skills, and abilities over broad hireability constructs, avoiding human-based training data, focusing on specific behavioral cues, and meticulously choosing interview questions. These principles collectively aim to mitigate biases in the selection process and define a more ethical path forward for this domain. Adhering to these principles will also assist organizations in developing systems that comply with evolving government regulations, like the EU's Artificial Intelligence Act and Illinois's Artificial Intelligence Video Interview Act. Such regulations could significantly limit the application of AI tools in selection processes, particularly those that use audio-video technologies, unless the systems are designed from the outset to ensure validity and fairness.

Further, our view is that systems like automated AVIs should be used to augment human decision-making, not completely replace it. For instance, assessing factors such as organizational fit is likely best left for human interpretation. As Rai, Constantinides, and Sarker point out, further studies on human-AI hybrid systems are necessary to grasp the broader consequences of this combined approach, particularly its impact on biased decision-making [115].

Our instantiation was only one minimal representative approach to predicting job performance rather than an optimal solution. While our results demonstrate that the STREAM-based instance performs above chance, chance is a relatively low benchmark, and the performance will need improvement to be acceptable for typical hiring contexts. There is a clear avenue for future research to enhance or add to the design principles in a manner that will improve predictive performance and reduce bias. For instance, future research should concentrate on the second stage of fusion shown in Fig. 2 to identify the optimal signal models and respective weightings for various industries and job types. Future research can also explore design (e.g., question delivery), behavioral (e.g., thought complexity), or cultural (e.g., non-native speakers) components that may enhance or detract from predictive ability and fairness. This includes how interviewees' reactions to automated AVIs—such as their perceptions of justice, levels of anxiety, and use of impression management strategies—may affect their behaviors as well as other interview outcomes [2]. It can also explore predicting factors related to performance, such as citizenship behaviors or counterproductive behaviors.

Additionally, our evaluation assumes the WPT is an objective measure of GMA. Though the WPT is more objective than human perception, it is only a proxy for intelligence. A participant's WPT score could vary depending on situational factors (e.g., being tired or hungry), which could potentially affect the classification. Further, existing standardized assessments, including GMA assessments, have been found to unfairly discriminate against minority groups[1] [116]. Future work should continue to strive for even more precise, reliable, and equitable measures of job performance. This exploration will likely extend to alternatives beyond traditional tools like written assessments for training automated systems, ensuring that emerging data-driven technologies are not constrained by the limitations of current methodologies.

Finally, while participants in our experiment were driven to excel in the Wonderlic assessment and during their interview responses, one must note that real-world employment interviews carry higher stakes than simulated hiring scenarios. Such genuine scenarios might escalate stress levels, potentially impacting voice and speech patterns. Moreover, our study was confined to a U.S. demographic. To gain a holistic understanding of language and voice characteristics predictive of the desired constructs, future research must cast a wider net, encompassing a broader demographic spectrum.

9. Conclusion

We set out to propose design principles and evaluate an automated AVI system with the potential to increase objectivity while preserving efficiency and costs. Given the rapid advancement of AI-based assessment systems and increasing government scrutiny, our work provides an important set of guidelines that can increase the efficacy and fairness of automated AVIs. This research underscores the potential for these systems to revolutionize the hiring landscape, provided they are developed with an emphasis on ethical standards and inclusivity, ensuring that the progress in AI contributes to a fairer and more equitable employment process for all.

[1] We follow the approach in [117] and evaluate the biases present in the WPT scores in our sample. The correlation between age and WPT scores is −0.170 ($p = 0.085$). Cohen's d metrics for ethnicity (White vs. non-White) and gender are 0.287 and 0.159, respectively. Because a correlation ≥ 0.10 or Cohen's $d \geq 0.20$ is considered indicative of bias, the WPT scores are slightly biased against older and non-White individuals but unbiased in terms of gender. We highlight this inherent bias in the WPT scores as a limitation and a direction for future research.

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Declaration of generative AI in scientific writing

In preparing this document, the authors utilized ChatGPT for enhancing text readability and conciseness. This tool was not used to produce any new ideas or original content. The authors thoroughly reviewed and revised the output as necessary, assuming full responsibility for the publication's content.

CRediT authorship contribution statement

Steven J. Pentland: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Xinran Wang:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Nathan W. Twyman:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization.

Declarations of competing interest

The authors declare no conflict of interest.

Appendix A—Summary Statistics and Correlation Matrix

Table A1, Table A2, Table A3

Table A1
Summary Statistics of Variables for Regression.

	Mean	Standard Deviation	Minimum	Maximum	Observations
WPT-Q score	27.729	4.609	17.000	36.000	413
Disfluency ratio	0.039	0.031	0	0.178	413
Filler words	0.034	0.027	0	0.151	413
Response duration in second	49.967	13.728	7.980	65.220	413
Speech rate	2.498	0.454	1.033	3.859	413
Cognitive processes	15.902	6.265	1.890	41.300	413
Lexical diversity	0.601	0.080	0.390	0.860	413
Average word length	4.012	0.280	3.217	4.938	413
SMOG	11.918	2.695	3.129	25.254	413
Hedge and uncertainty	18.395	10.308	0	66.000	413
Expressivity	0.096	0.049	0	0.400	413
Pitch variability	125.980	25.970	71.959	186.009	413
Amplitude variability	0.712	0.067	0.481	0.875	413

Table A2
Correlation Matrix of Variables for Regression.

		1	2	3	4	5	6	7	8	9	10	11	12	13
1	WPT-Q score	1	−0.161	−0.124	0.330	0.204	0.033	−0.111	0.182	0.111	0.296	−0.298	−0.044	0.058
2	Disfluency ratio		1	0.932	0.023	−0.239	−0.122	−0.090	−0.182	0.022	−0.116	0.029	0.093	−0.075
3	Filler words			1	0.071	−0.214	−0.144	−0.094	−0.152	0.031	−0.080	−0.018	0.110	−0.108
4	Response duration in second				1	0.156	−0.215	−0.605	0.177	0.152	0.488	−0.763	−0.129	0.045
5	Speech rate					1	0.081	−0.480	−0.130	−0.040	0.532	−0.350	−0.084	0.072
6	Cognitive processes						1	0.071	−0.120	0.036	0.292	0.315	−0.016	0.055
7	Lexical diversity							1	0.247	−0.009	−0.475	0.521	0.156	−0.040
8	Average word length								1	0.428	0.023	−0.258	−0.028	−0.050
9	SMOG									1	0.147	−0.068	−0.039	−0.037
10	Hedge and uncertainty										1	−0.407	−0.093	0.057
11	Expressivity											1	0.116	−0.020
12	Pitch variability												1	−0.067
13	Amplitude variability													1

Appendix B—Vocalic and Linguistic Modality Comparison

To compare the predictive power of the vocalic and linguistic modalities, we further construct a dataset with 74 vocalic features and another dataset with 85 linguistic features. We perform the same model training, hyperparameter tuning, and evaluation procedures on the same 50 random splits with these two unimodal datasets. Panels A and B of Table A3 report the performances using the vocalic and linguistic datasets, respectively. The benchmark performances are those obtained using the multimodal dataset reported in Table 7. Paired *t*-tests are conducted to compare model performances of the multimodal and unimodal datasets. Using the vocalic features alone outperforms random guess baselines, indicating that vocalic features are predictive of GMA. However, compared with using the multimodal dataset, most of the performance metrics significantly decrease. In contrast, using only linguistic features does not harm performances. These results on our dataset imply that although vocalic features entail discriminative ability for GMA prediction, linguistic features are more telling. Given a small to medium-sized sample, when vocalic features are added to linguistic features, model performances saturate.

Table A3
Performance of GMA Prediction When Using Only the Vocalic (Panel A) or Linguistic (Panel B) Features.

	(A) Performance—Voice Features Only (Change from Multimodal Model)			(B) Performance —Language Features Only (Change from Multimodal Model)		
Classifier	Accuracy (Δ Accuracy)	F1 (Δ F1)	AUC (Δ AUC)	Accuracy (Δ Accuracy)	F1 (Δ F1)	AUC (Δ AUC)
LR	0.635 (−0.069)**	0.624 (−0.064)*	0.699 (−0.060)*	0.677 (−0.027)	0.660 (−0.028)	0.732 (−0.027)
SVM	0.641 (−0.051)*	0.609 (−0.051)	0.669 (−0.088)***	0.696 (0.004)	0.685 (0.025)	0.752 (−0.005)
RF	0.613 (−0.083)***	0.614 (−0.075)***	0.657 (−0.099)***	0.708 (0.012)	0.703 (0.014)	0.755 (−0.001)
AB	0.655 (−0.042)*	0.654 (−0.035)	0.703 (−0.058)*	0.697 (0.000)	0.678 (−0.011)	0.768 (0.007)
BAG	0.640 (−0.039)	0.630 (−0.030)	0.689 (−0.052)*	0.689 (0.010)	0.667 (0.007)	0.736 (−0.005)

Note: The benchmark performances are those reported in Table 7. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

Appendix C—Fairness of GMA Classification

Evaluating and mitigating biases in hiring decision-support systems is essential for promoting social justice and enhancing public trust in these technologies. Thus, we conduct further analyses to assess the fairness of our algorithmic predictions. Specifically, we investigate whether our system discriminates against historically unprivileged groups, including females, individuals above the median age, and people of color. Because only nine participants were non-native English speakers, we do not examine biases related to native languages, but future research should address this limitation by using a more diverse participant pool.

We denote high (or low) GMA labels to be $GMA = 1$ (or $GMA = 0$) and high (or low) GMA predictions to be $\widehat{GMA} = 1$ (or $\widehat{GMA} = 0$). We adopt the fairness criteria of equal opportunity [1] and disparate impact [2] and define the following metrics to evaluate algorithmic fairness.

$$EOD_{gender} = P(\widehat{GMA} = 1 | GMA = 1,\ Female) - P(\widehat{GMA} = 1 | GMA = 1,\ Male)$$

$$EOD_{age} = P(\widehat{GMA} = 1 | GMA = 1, Age\ \geq median\ ) - P(\widehat{GMA} = 1 | GMA = 1,\ Age\ < median)$$

$$EOD_{ethnicity} = P(\widehat{GMA} = 1 | GMA = 1, People\ of\ color) - P(\widehat{GMA} = 1 | GMA = 1, Whites)$$

$$DI_{gender} = \frac{P(\widehat{GMA} = 1 | Female)}{P(\widehat{GMA} = 1 | Male)}$$

$$DI_{age} = \frac{P(\widehat{GMA} = 1 | Age\ \geq median)}{P(\widehat{GMA} = 1 | Age\ < median)}$$

$$DI_{ethnicity} = \frac{P(\widehat{GMA} = 1 | Peope\ of\ color)}{P(\widehat{GMA} = 1 | Whites)}$$

The disparate impact measure aligns with the guidelines set forth by the U.S. Equal Employment Opportunity Commission [3], which advocates for adherence to the 80 % rule. This rule states that protected groups should be hired at a rate no less than 80 % of the rate at which non-protected groups are hired. Thus, DI_{gender}, DI_{age}, and $DI_{ethnicity}$ should be no less than 80 % per the guidelines.

We evaluate the fairness metrics by comparing the ground truth labels with the model predictions across all the test sets. Results in Tables A4, A5 and A6 show that the proposed system generates fair predictions with respect to gender, age, and ethnicity, which adhere to the 80 % rule.

Table A4
Fairness of GMA Classification by Gender.

Classifier	Equal Opportunity Difference (EOD)	Disparate Impact (DI)
LR	0.021	1.100
SVM	0.037	1.100
RF	0.085	1.087
AB	0.005	1.038
BAG	0.002	1.049

Table A5
Fairness of GMA Classification by Age.

Classifier	Equal Opportunity Difference (EOD)	Disparate Impact (DI)
LR	−0.006	0.953
SVM	0.040	0.990
RF	0.088	1.068
AB	0.022	0.979
BAG	0.002	0.958

Table A6
Fairness of GMA Classification by Ethnicity.

Classifier	Equal Opportunity Difference (EOD)	Disparate Impact (DI)
LR	−0.020	1.081
SVM	−0.050	1.064
RF	−0.062	1.087
AB	−0.054	1.047
BAG	−0.039	0.988

Appendix D—Performance of GMA Regression

In addition to GMA classification, we also formulate GMA prediction as a regression task. We employ the same 70 %−15 %−15 % splits for model training, validation, and testing as in the classification task. We use the same set of 74 vocalic features and 85 linguistic features as classification to train a suite of widely adopted machine learning models, including ridge, lasso, elastic net, support vector, and random forest regression. We do not apply additional variable selection because regressors such as lasso and elastic net regression inherently incorporate a regularization mechanism that includes variable selection. For hyperparameter tuning, we average the root mean squared error (RMSE) on all the validation sets and select the optimal set of hyperparameters that achieves the lowest RMSE. We report the average Pearson correlation coefficient (R), RMSE, and mean absolute error (MAE) across the test sets in Table A7. The random guess baseline predictions are generated by randomly selecting a WPT-Q score from the training set. Another naïve baseline uses the average WPT-Q scores from the training set for prediction. Paired *t*-tests indicate that all regressors significantly outperform both baselines across all metrics ($p < 0.01$). The average R across all regressors is 0.406, showing a moderate correlation between our predictions and the WPT-Q scores. The average MAE of 3.623 suggests a relatively small average absolute error, given the average range of WPT-Q scores across all test sets is 15.06.

We further compare RMSE with the standard deviations of WPT-Q scores across all test sets. This stringent benchmark assumes predictions are the average WPT-Q score of the test sets (unknown during model training) and represents the natural variability of the predicted values. The average standard deviation of WPT-Q scores across all test sets is 4.472, marginally above the regressors' RMSEs. Paired *t*-tests reveal that all regressors significantly outperform this more stringent benchmark ($p < 0.05$).

The regression results demonstrate the viability of our system to predict GMA scores as a continuous variable. Our system facilitates more accurate predictions of job candidates' GMA levels and enables organizations to select candidates more effectively.

Table A7
Performance of GMA Regression.

Performance – Multimodal Model			
Regressor	**R**	**RMSE**	**MAE**
Ridge	0.401	4.306	3.610
Lasso	**0.429**	4.335	3.662
Elastic Net	0.421	**4.243**	**3.554**
Support Vector	0.385	4.316	3.638
Random Forest	0.394	4.323	3.653
Random Guess Baseline	0.021	6.387	5.176
Average Training GMA Baseline	0	4.677	3.991

Note: The best performance metrics are in bold.

Appendix E—Candidate Ranking Evaluation: Comparison of *T* Values

In addition to ranking the top 50 % of participants based on GMA scores, the table below reports performance results when predicting the top 5 and top 3 participants. Results should be interpreted with respect to the random classification baseline rate.

	Performance—Multimodal Model T = Number of top candidates to be identified					
	T = 50 % of test set		T = 5		T = 3	
Classifier	**Precision**	**Recall**	**Precision**	**Recall**	**Precision**	**Recall**
Random Classification Baseline	0.570	0.533	0.357	0.314	0.219	0.183
LR	0.733	0.685	**0.612**	0.543	**0.560**	**0.472**
SVM	0.715	0.667	0.600	0.535	0.547	0.463
RF	0.720	0.672	**0.612**	**0.547**	0.520	0.433
AB	**0.735**	**0.687**	0.600	0.533	0.553	0.467
BAG	0.702	0.656	0.608	0.545	0.540	0.459

Note: The best performance metrics are in bold.

Appendix F—Question Type Evaluation

To evaluate the predictability of different question types, we replicate the classification with features extracted from each question type (i.e., PBQs or SQs). We adopt the same model training, hyperparameter tuning, and evaluation procedures on the same 50 random splits. Fig. A1 compares the performance metrics of each question type with those of the full dataset (full dataset performance also reported in Table 7). We also conduct paired *t*-tests to test the statistical significance of the differences in performances. Using only the PBQs generally results in similar performances to using the full dataset, and the differences are insignificant at 0.05 level for LR, SVM, AB, and BAG. In contrast, using only the SQs generally results in lower performances than using the full dataset, and the differences are mostly significant at 0.05 level. The only exception is that SQs and the full dataset do not differ significantly in the F1 score when SVM is used. One possible explanation for the comparison is that PBQs are more telling of GMA than SQs. Because the order of questions remains the same in our experiment, another explanation could be that GMA is revealed more in early questions than in late questions.

To explore whether the question order or types are the more probable reason for changes in predictive performances, we replicate the classification with features extracted from each question and plot Fig. A2. We also conduct paired *t*-tests to compare Q1 and Q2, Q2 and Q3, and Q3 and Q4. Q1 and Q2 do not differ significantly in most of the performance metrics across the five classifiers at 0.05 level. The exceptions are that accuracy with LR and SVM and AUC with LR and BAG increase ($p < 0.05$) from Q1 to Q2. In contrast, many performance metrics decrease significantly at 0.05 level from Q2 to Q3. Specifically, all the three metrics decrease when LR, SVM, and BAG are used, and AUC decreases when AB is used. Q3 and Q4 do not differ significantly in any performance metrics at 0.05 level. Because Q1 and Q2 are both PBQs, and Q3 and Q4 are both SQs, the results provide preliminary evidence that PBQs are more telling of GMA than SQs. Future research is needed to confirm or disconfirm this exploratory analysis.

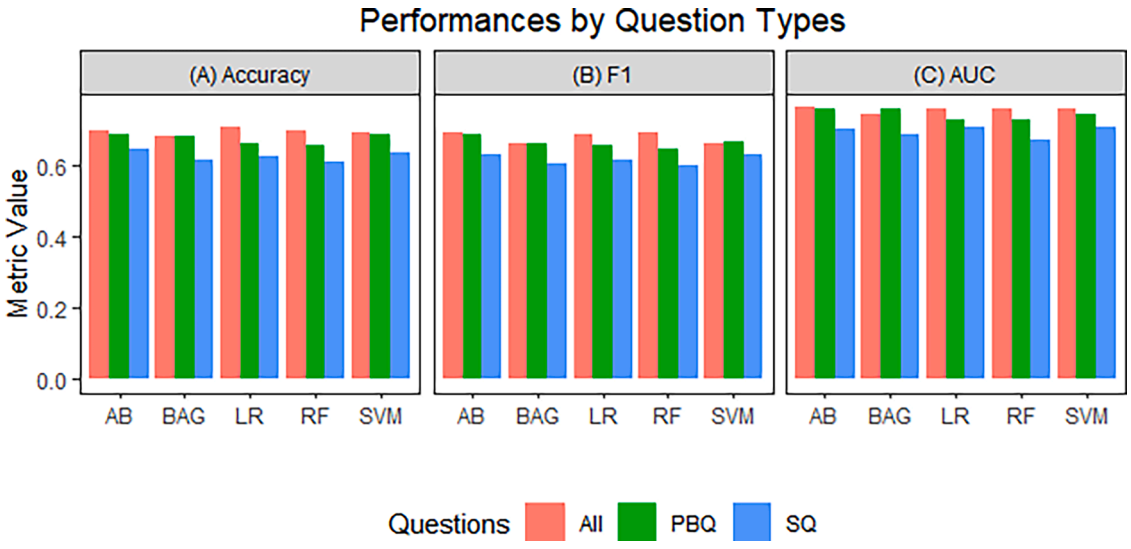


Fig. A1. Performance of GMA Prediction by Question Type.

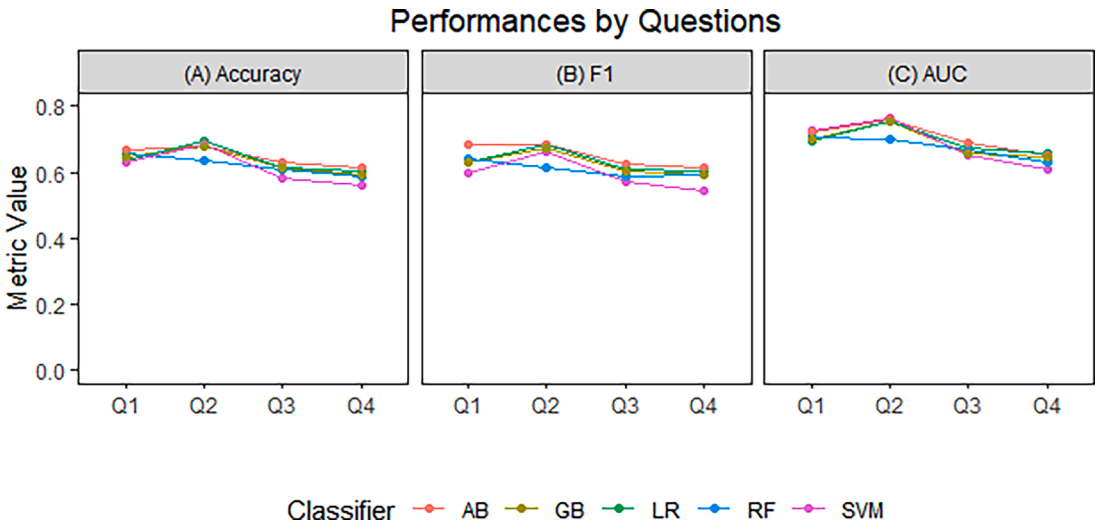


Fig. A2. Performance of GMA Prediction by Question.

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